DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #04A: WEAPONS ARSENAL (PART 1 — CORE BLADED & POLEARM MELEE FAMILIES: SWORD, BLADE, SPEAR, STAFF)

- **Target Codebase**: JianghuSect 2 (JS2-v2.6.0 / SIM.PATCH_VERSION = '2.6.0' / HC.FINGERPRINT = '155:2f9c9dd7')
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Weapon Audit Series Plan (04A–04E Granular 16-Family + Holistic Ecosystem Audit)**:
  - **04A (This Document)**: Group 1 — **Core Bladed & Polearm Melee (SWORD กระบี่, BLADE ดาบ/ดาบใหญ่/มีดสั้น, SPEAR ทวน/หอก/ง้าว, STAFF พลอง/กระบอง/ค้อน)**
  - 04B: Group 2 — Brawling, Ballistic & Flexible (FIST, CLAW, BOW, NEEDLE, WHIP)
  - 04C: Group 3 — Scholarly, Musical & Dharma Conduits (FOCUS, ZITHER, BRUSH, ROSARY)
  - 04D: Group 4 — Grand Dao Tactical Implements (FAN, ASTROLABE, ABACUS)
  - 04E: Part 5 — Holistic Weapon Ecosystem & Catch-All Cross-System Audit (Classification Regex, Procedural Forge, Soul-Forge Armament Wills, Runes, Durability/Market Sink, Unarmed Fallback & 51 Legendary Relics)
- *(Cross-checked against JS2_v2.4.0_Deep_QA_Equipment_Report_TH.docx & JS2_V2_4_0_FIX_TRACKER_TH.md so previously closed v2.4.0 bugs are excluded; every defect below is verified live in JS2-v2.6.0.)*

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable human_core.js**: Never modify GAME_SOURCE/js/human_core.js (HC.FINGERPRINT === '155:2f9c9dd7'). Note that h.skills natively holds the 7 core keys SWORD, BLADE, SPEAR, FIST_PALM, BOW, STAFF, FOCUS.
2. **40-Script Load Order & Zero Render RNG**: Preserve exact <script> order in index.html; Render weapon silhouettes and WEAPON_MOTION_VFX must never advance W.rng or mutate game state.
3. **100% Emergent NPC Autonomy**: NPCs evaluate, craft, equip, maintain, and fight with weapons organically based on subpath, physique (strength, agility, spiritualFoundation), and tactical spacing—without player micromanagement.

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1. [WPN-SWORD-01] Weapon Family #1: SWORD (กระบี่ — Swift Jians, Heavy Greatswords & Telekinetic Flying Swords)

1.1 Heavy Greatsword Tactical Misclassification (LEG_WPN_XUANTIE_HEAVY_SWORD — กระบี่เหล็กนิลไร้คม)

- **Locations**: `js/visitors_clues.js:55–79`, `js/skills.js` (`SK.WEAPON_TACTICS`), `js/combat.js` (`resolveWeaponTactics`)
- **Contextual & Tactical Contradiction**:
  - `LEG_WPN_XUANTIE_HEAVY_SWORD` (玄铁重剑 — Yang Guo's 81-catty blunt Dark-Iron Heavy Sword) explicitly defines `visualSubtype: 'HEAVY_SWORD'`, `reqStrength: 58`, `move: -8`, `cast: -6`, and lore: "ปลายมนไร้คม อาศัยพละกำลังมหาศาลบดขยี้ทุกกระบวนท่า".
  - **Bug**: Because `resolveWeaponTactics` inspects only `gearType === 'SWORD'` and ignores `visualSubtype === 'HEAVY_SWORD'`, `XUANTIE_HEAVY_SWORD` is treated as **`SHORT_SWIFT`** (**อาวุธเบาพลิ้วคลุกวงใน**) —so it **cannot shatter enemy `guardUp`/shields** (`shatterGuard: false`) and actually **gets countered by ordinary `BLADE`s (`HEAVY_BREAKER`)**!
- **Fix**: In `SK.weaponTacticalProfile(u)`, when `wpn.visualSubtype === 'HEAVY_SWORD'` (or `wpn.reqStrength >= 55`), override `reachClass` to **`'HEAVY_BREAKER'`** (`shatterGuard: true`, `+14 pen` vs guard/polearms)!

1.2 Dead `armorReductionRatio` Parameter on `YITIAN_SWORD` & `XUANTIE_HEAVY_SWORD` (`visitors_clues.js:47, 76, 682–695, 754–783`)

- **Exact Code Proof**:
  - `LEG_WPN_YITIAN_SWORD` sets `armorReductionRatio: 0.25` and `LEG_WPN_XUANTIE_HEAVY_SWORD` sets `armorReductionRatio: 0.30` inside `effects[0].params`.
  - In `visitors_clues.js:682–695`, the `ST.gearStats` wrapper returns `def.fixedStats` **early** without running `origGearStats`'s `s.pen += p.armorReductionRatio * 40` (`stats.js:68`), **and** in `VC.runWeaponProcs` (`lines 754–783`), `p.armorReductionRatio` is never read (`XUANTIE` only applies `'slow'`, never `'armorbreak'` or bonus armor ignore)!
- **Second Bug on `LEG_WPN_YITIAN_SWORD` (กระบี่อิงฟ้าพิฆาตมาร, `lines 46, 754–773`)**:
  - Tooltip promises: "เมื่อโจมตีเป้าหมายที่มีโล่**หรือตั้งรับ *(`guardUp`)*** จะทำลายโล่ *35%* ทันที..."

- `EFX_LEG_YITIAN_SEVER` cuts `shield` / `shieldBp` / `defender.shield`, **but completely forgets to check or strip `ST.has(defender, 'guardUp')`**, and burns its 8-tick ICD even against targets with zero shield and zero guard!
- **Fix**: Have `EFX_LEG_YITIAN_SEVER` also strip `guardUp` (dealing `+15%` guard-piercing sever bonus when stripping `shield` or `guardUp`), and apply `p.armorReductionRatio` as temporary defense reduction on `EFX_LEG_XUANTIE_CRUSH`!

1.3 Telekinetic Flying Sword Melee Lock (`MIND_SWORD_TELEKINESIS` & `FLYING_BLADE_ORBITER`)

- Disciples on `HEART_SOUL.MIND_SWORD_TELEKINESIS` (สายจอมกระบี่จิตไร้รูป) or `MECH_PUPPET.FLYING_BLADE_ORBITER` (กงล้อศาสตราพันกระบี่) control hovering swords via mental Qi, yet basic attacks and default `SWORD` tactics lock them to `1.80`-tile melee range (`SHORT_SWIFT`), forcing squishy `HEART_SOUL` scholars (`def * 0.94, hp -6%`) to run into melee!
- **Fix**: When `c.sub` is `MIND_SWORD_TELEKINESIS`, `FLYING_BLADE_ORBITER`, or `CELESTIAL_STAR_SWORD`, elevate `SWORD` basic/tactical reach to `MID_TACTICAL` (`baseRange: 2.55`, `RANGED_KITE` projection)!

1.4 Hardcoded `'SWORD'` Practice Leak on Clue Guardian Victories (`visitors_clues.js:1798`)

- **Exact Code (`visitors_clues.js:1798`)**:

```javascript
(M && M.skillPractice ? M.skillPractice : HC.practice)(m, 'SWORD', 1.2, 1.1);
```

- **Bug**: Every party member who wins a Clue Guardian fight practices `'SWORD'`—even pure `SPEAR`, `STAFF`, `BLADE`, `FIST_PALM`, or `ZITHER` cultivators! Replace `'SWORD'` with `M.equippedSkillKey(m)` (mapping `'FIST'` $\to$ `'FIST_PALM'`).

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2. [WPN-BLADE-02] Weapon Family #2: BLADE (ดาบ / ดาบใหญ่ / มีดสั้น / กริช — Four Conflicting Archetypes in One Bucket)

2.1 Critical Triple Bug on LEG_WPN_XIAOLI_HEART_DAGGER (มีดสั้นเสี่ยวหลี่สะท้านภพ — 小李飞刀)

Inspecting `visitors_clues.js:129–158`, `804–816`, and `948–955` reveals **three severe logical and code bugs** on one of Wuxia's most iconic weapons:

1. **Zero-Qi `SHADOW_SAMURAI` Permanent Proc & True-Strike Lockout (`lines 805–807, 951`)!**:

```javascript
const qm = CULT.qiMax(attacker);
const needQi = Math.max(1, Math.round(qm * (p.qiCostPct || 0.04)));
if ((attacker.h.state.qi || 0) >= needQi) { ... }
```

   - `SHADOW_SAMURAI` is strictly restricted to `SWORD` and `BLADE` (`XIAOLI_HEART_DAGGER` is `gearType: 'BLADE'`) and has **`CULT.qiMax(attacker) === 0`** (`h.state.qi === 0`).
   - Because `Math.max(1, 0)` evaluates to **`1`**, `0 >= 1` is **permanently `false`**! Neither the guaranteed hit (`oMod.forced = true` at `line 951`) nor `EFX_LEG_XIAOLI_TRUE_STRIKE` can **ever** fire for a `SHADOW_SAMURAI`!
   - **Fix**: When `CULT.isZeroQiSamurai(attacker)`, pay `needFocus = 8` from `IAI_FOCUS` (or `needEnergy = 3` from `h.state.energy`) instead of `qi`!

2. **Throwing Dagger Forced Into `1.75`-Tile Heavy-Greatsword Melee (`visualSubtype: 'DAGGER'` vs `gearType: 'BLADE'`)**:

   - *Little Li Flying Dagger* (`"ซัดมีดบินที่ไม่อาจหลบหลีกได้"`) is a **ranged throwing blade**, yet because `gearType === 'BLADE'`, `SK.WEAPON_TACTICS` assigns it `HEAVY_BREAKER` (`baseRange: 1.75` melee broadsword!), forcing the wielder to walk into melee fist-range before throwing a flying dagger!
   - **And** its #1 `preferredSubs` entry is `'MIND_SWORD_TELEKINESIS'` (`visitors_clues.js:145`), while `M.PROFILES.MIND_SWORD_TELEKINESIS` (`martial.js:55`) only prefers `['SWORD', 'FOCUS']` (penalizing `BLADE` suitability in `prevGearSuitability`)!

- **Fix**: Give `visualSubtype === 'DAGGER'` (and daggers like `ITEM190_EQ_006` กริชเขี้ยวแมงป่องเจ็ดพิษ) a sub-tactical profile: throwing daggers (`XIAOLI`) use `RANGED_KITE` (`baseRange: 2.75`), assassin daggers (กริช / `FULL_MOON_SCIMITAR`) use `SHORT_SWIFT` (`+crit / +eva`), while heavy sabers (`TULONG_SABER` / `OVERLORD_HEAVY_BLADE`) retain `HEAVY_BREAKER`!

3. **Dead `bonusPen: 22` Parameter in `EFX_LEG_XIAOLI_TRUE_STRIKE` (`line 153` vs `804–816`)**:

    - `params` declares `bonusPen: 22` (*"ทะลวงเกราะ +22%"*), **but `COMBAT.strike` (`line 953`) runs `origStrike` BEFORE `VC.runWeaponProcs`, and `oMod` at `line 951` only sets `oMod.forced = true`, completely forgetting `oMod.pen = (oMod.pen || 0) + 22`!**

2.2 All Pure `BLADE` Subpaths Missing From `M.SUB_EQUIP_BONUSES` (`js/martial.js:465–481`)

- `M.SUB_EQUIP_BONUSES` defines custom weapon bonuses for `SWORD_IMMORTAL` and `PURE_SWORD_SAINT`, **but omits every single pure `BLADE` subpath in the game** (`OVERLORD_HEAVY_BLADE`, `ASHURA_SLAUGHTER_BLADE`, `ASURA_JUDGMENT_BLADE`, `CRIMSON_MOON_EXECUTIONER`, `HEART_BLADE_MEDITATION`, `IAI_FLASH_SLICER`, `MURAMASA_BLOOD_EDGE`, `DUELIST_HONOR_BLADE`), dumping them into the generic `{ atk: +6%, matk: +6% }` fallback (wasting half the bonus on `matk` for pure physical sabers)!

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3. `[WPN-SPEAR-03]` Weapon Family #3: `SPEAR` (`ทวน / หอก / ง้าว` — Polearm Outreach & Vanguard ZoC)

3.1 Three Concrete Bugs in `SPEAR` Combat & Relic Execution

1. **Enemy/Rival `SPEAR` Zone-of-Control (`zocSlowMult`) Bypassed by Sect NPCs (`js/combat.js:232–255`)**:
    - `SPEAR` vanguards project `unitZoCRadius` to slow chargers (`0.52×` speed) and protect backline archers/healers (`SPEAR_ZOC_COUNTER`). Because `!u.foe` skips `zocSlowMult` in `COMBAT.moveToward`, sect assassins sprint past enemy spear walls unhindered in duels, tournaments, and sect wars!
2. **`LEG_WPN_OVERLORD_SPEAR` (ทวนราชันย์พิชิตพยัคฆ์) Broken Trigger & Wasted Cooldown (`visitors_clues.js:186, 817–827`)**:

- Tooltip states: *"ยิ่งผู้ถือมีความกล้าหาญสูง**หรือสู้กับศัตรูที่แข็งแกร่งกว่า** จะเพิ่มความแรง *+15%* และฟื้นฟู**แรงกาย**/ปราณ *4%"*.
- **Bugs in `lines 817–827`**:
  - Sets cooldown `attacker.wpnProcCds[eid] = W.t + icd` **unconditionally on line 818** even when `courage < 55` (wasting the proc for 0 damage!).
  - Completely omits the `|| defenderPower > attackerPower` underdog-courage check (which is the literal literary theme of Gu Long's *Overlord Spear*!).
  - Restores only `qi`, forgetting `energy` (แรงกาย), and ignores `params.armorReductionRatio: 0.22`!

3. **Missing Mounted Polearm Synergy (`MOUNT_CHARGER` + `SPEAR`)**:
   - Grant `+12%` Round-1 Opening Charge damage & poise break when a `SPEAR` wielder fights alongside an active `MOUNT_CHARGER` / `UNGULATE` spirit beast!

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4. `[WPN-STAFF-04]` Weapon Family #4: `STAFF` (`พลอง / กระบอง / ค้อนศึก` — Redirection Staffs vs Blunt Warhammers)

4.1 Live Copy-Paste Bug in `VC.runWeaponOnStruckProcs` (`visitors_clues.js:918–940`)

- **Exact Code (`visitors_clues.js:919–938`)**:

```javascript
if (fx.effectId === 'EFX_LEG_DAGOU_REDIRECTION' || fx.effectId ===
'EFX_LEG_KIRIN_PHANTOM_COUNTER' || fx.effectId ===
'EFX_LEG_XUANWU_UNYIELDING_BASTION' || fx.effectId ===
'EFX_LEG_CRANE_CLOUD_AEGIS') {
  ...
  COMBAT.applyDamage(W, f, defender, attacker, redirDmg, 'พลองหยกยืมแรงปัดสวน');
  ST.addStatus(attacker, p.targetStatus || 'slow', p.dur || 7);
  ST.addStatus(defender, p.selfStatus || 'evadeUp', p.dur || 7);
}
```

- **Four Compounded Bugs**:
  1. **Wrong Combat Text & Wrong Mechanics on All 3 Legendary Armors**: Wearing `LEG_ARM_CELESTIAL_CRANE_ROBE` or `LEG_ARM_IMMORTAL_TORTOISE_PLATE` prints `"พลองหยกยืมแรงปัดสวน"` **(Dagou Jade Staff!)**, deals phantom `15%` staff-redirection damage on Crane Robe (which has no reflect in its spec!), **fails to restore `5%` Qi (`qiRestorePct:`**

0.05) on Crane Robe, and **fails to cleanse armorbreak/stun on Tortoise Plate**!

2. **25-Tile Ranged Leg-Sweep Absurdity on LEG_WPN_DAGOU_JADE_STAFF (พลองหยกเขียวตีสุนัข)**: If shot by a distant archer (`dist > 2.6`), `DAGOU_JADE_STAFF` still sweeps the archer's legs (เกี่ยวสกัดขา) and strikes back across the map! Restrict physical leg-sweep counter-damage to melee attackers (`unitDist <= 2.6`), while granting projectile deflection (`evadeUp`) against ranged hits!
3. **Blunt Warhammers (ค้อน — e.g. ITEM190_EQ_005 ค้อนสายฟ้าม่วงเหนี่ยวนำ)**: Ensure `JD.MARTIAL.classify('WEAPON', nameTh)` maps `/ค้อน|กระบอง|ลูกตุ้ม/` explicitly to `'STAFF'` with `visualSubtype: 'HAMMER'` (`HEAVY_BREAKER` guard-shatter, rather than falling through to `'SWORD'` in `classify`!).
4. **Broken Baseline Item IDs in VC.benchmarkWeaponBalance (visitors_clues.js:2112–2115)**: `STAFF` is benchmarked against `'MP_SPEAR_5'`, `BOW` against `'MP_SWORD_5'`, and `FOCUS` against `'MP_RING_5'` (a Ring accessory!) instead of `'MP_STAFF_5'`, `'MP_BOW_5'`, and `'MP_FOCUS_5'`!

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5. [WPN-DUR-05] Non-Procedural Weapon Decay Trap: Authored & Legendary Melee Weapons Rotting to 30% Power Forever!

- **Locations**: `js/sim.js` (`SIM.hourly`), `js/economy.js` (`ECO.wearGear`), `js/stats.js:368, 600`
- **Chain-Reaction Bug**:
  1. `SIM.hourly` applies `-0.5` durability wear per hour in combat to **all** equipped gear, driving authored (`ITEM190_EQ_*`, `MP_*`) and Legendary (`LEG_WPN_*`, `REL_V230_*`) weapons down to `dur = 1`.
  2. `stats.js:368, 600` multiplies every weapon stat by `df = clamp(it.dur / 100, 0.3, 1)`—so at `dur = 1`, **Excalibur-tier Legendary Weapons lose -70% of all stats (940 ATK $\to$ 282 ATK)**!
  3. Meanwhile, autonomous NPC maintenance in `SIM.hourly` checks `if (it.procedural)`—so **NPCs NEVER repair authored or Legendary weapons**, and if repaired via `ECO.repairGear`, `maxDur` would permanently shrink (`100 -> 80 -> ... -> 20`) on irreplaceable Unique World Artifacts!
- **Fix**: Exempt `it.legendary || it.unique` relics from permanent `maxDur` loss (`spirit-nourishment polish` restores `dur` to `100` during meditation/rest or via `SOUL_FORGE`), and allow NPCs to maintain non-procedural authored weapons with Whetstones without destroying `maxDur`!

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6. Implementation & Verification Checklist for GPT / Gemini (04A)

- [ ] **SWORD Fixes**: Map `visualSubtype === 'HEAVY_SWORD'` (`XUANTIE`) to `HEAVY_BREAKER`; make `EFX_LEG_YITIAN_SEVER` strip `guardUp` and only trigger ICD when useful; apply `armorReductionRatio` on Legendary weapon procs; grant `MID_TACTICAL` flying-sword reach to `MIND_SWORD_TELEKINESIS`/`FLYING_BLADE_ORBITER`; fix hardcoded `'SWORD'` practice in `simulateGuardianEncounter` (`visitors_clues.js:1798`).
- [ ] **BLADE Fixes**: Fix `XIAOLI_HEART_DAGGER` zero-Qi lockout on `SHADOW_SAMURAI` (`use IAI_FOCUS / energy`), apply its missing `bonusPen: 22` in `COMBAT.strike`, and give `DAGGER` (`RANGED_KITE` for throwing daggers / `SHORT_SWIFT` for assassin daggers) accurate sub-tactics; add pure `BLADE` subpaths to `M.SUB_EQUIP_BONUSES`.
- [ ] **SPEAR & STAFF Fixes**: Enforce `SPEAR` `zocSlowMult` on `!u.foe` units in `COMBAT.moveToward`; fix `OVERLORD_SPEAR` `defenderPower > attackerPower` check, `energy` restore, and conditional ICD; split `runWeaponOnStruckProcs` (`visitors_clues.js:918–940`) so `DAGOU_JADE_STAFF` checks melee range and the 3 Legendary Armors use their own names, Qi restore, and stun/armorbreak cleanse; map `/ค้อน|กระบอง/` to `STAFF` (`HAMMER` `HEAVY_BREAKER`); fix `benchmarkWeaponBalance` baseline IDs; and fix non-procedural/Legendary durability maintenance.